

Past Paper SPRING 2024

DAY: _____ CODE = PHYS-3304

$\Rightarrow$ Quantum Mechanics-I

Question # 1:-

Answer the following short Questions.

~(i)~.

Prove that eigen value of Hermitian operator is real?

~(Answer)~.

Example # 2.15 (Chap 2)

Page # 65, 66.

Imi Quantum Mechanics.

~(ii)~.

Define wave function. What are properties of a wave function?

~(Answer)~.

Wave Function:-

A wave-function usually

denoted by $\psi(x,t)$, is a complex-valued function that contains all the information about a quantum system. The probability of finding a particle at a position ' x ' and time ' t ' is given by $|\psi(x,t)|^2$.

Properties of a wave-function:-

1. **Single valued:** $\psi(x,t)$ must have only one value for a given x and t .
2. **Continuous:** $\psi(x,t)$ must be continuous and differentiable (unless there is an infinite potential).
3. **Normalizable:** The integral over all space $\int |\psi(x,t)|^2 dx = 1$, so that the total

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probability is 1.

4. **Finite:** The wave-function must be finite everywhere.

5. **Square-integrable:** $\psi(x, t)$ must belong to L^2 space, i.e.; its square modulus must be integrable.

~(iii)~.

Find out if the function $\psi(x) = e^{ax}$ is an eigen-function of x -component of momentum operator.

~(Answer)~.

The x -component of the momentum operator in quantum mechanics is:

$$\hat{p}_x = -i\hbar \frac{d}{dx}$$

Apply $\hat{p}_x$ to $\psi(x) = e^{ax}$;

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$$\hat{p}_x \psi(x) = -i\hbar \frac{d}{dx} (e^{ax})$$

$$= -i\hbar a e^{ax}$$

We say that;

$$\hat{p}_x \psi(x) = \lambda \psi(x)$$

as;

$$\lambda = -i\hbar a \quad \text{and}$$

$$\psi(x) = e^{ax} \quad (\text{given}).$$

conclusion:-

Yes, $\psi(x) = e^{ax}$ is an eigenfunction of momentum operator $\hat{p}_x$ with eigen value $-i\hbar a$.

Given:-

Consider two states

$$|\psi_1\rangle = 2i|\phi_1\rangle + |\phi_2\rangle - \alpha|\phi_3\rangle + 4|\phi_4\rangle.$$

$$|\psi_2\rangle = 3|\phi_1\rangle - i|\phi_2\rangle + 5|\phi_3\rangle - |\phi_4\rangle$$

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where $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle$
and $|\phi_4\rangle$ are orthonormal
kets and α is constant.

Find value of ' α ', if
 $|\psi_1\rangle$ and $|\psi_2\rangle$ are
orthogonals.

Answer :-

By the definition of
orthogonality;

$$|\psi_1\rangle |\psi_2\rangle \rightarrow \text{given}$$

$$\langle \psi_1 | \psi_2 \rangle = 0$$

$$\langle \psi_1 | = (|\psi_1\rangle)^*$$

$$\langle \psi_1 | = [2i|\phi_1\rangle + |\phi_2\rangle - \alpha|\phi_3\rangle + 4|\phi_4\rangle]^*$$

$$\langle \psi_1 | = [2i\langle \phi_1 | + \langle \phi_2 | - \alpha\langle \phi_3 | + 4\langle \phi_4 |]$$

$$\langle \psi_1 | \psi_2 \rangle = [2i\langle \phi_1 | + \langle \phi_2 | - \alpha\langle \phi_3 | + 4\langle \phi_4 |]$$

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$$[3|\phi_1\rangle - i|\phi_2\rangle + 5|\phi_3\rangle - |\phi_4\rangle]$$

$$= -6i\langle\phi_1|\phi_1\rangle + 2i^2\langle\phi_1|\phi_2\rangle - 10i\langle\phi_1|\phi_3\rangle + 2i\langle\phi_1|\phi_4\rangle + 3\langle\phi_2|\phi_1\rangle - i\langle\phi_2|\phi_2\rangle - 5\langle\phi_2|\phi_3\rangle - \langle\phi_2|\phi_4\rangle - 3\alpha\langle\phi_3|\phi_1\rangle + \alpha i\langle\phi_3|\phi_2\rangle - 5\alpha\langle\phi_3|\phi_3\rangle + \alpha\langle\phi_3|\phi_4\rangle + 12\langle\phi_4|\phi_1\rangle - 4i\langle\phi_4|\phi_2\rangle + 20\langle\phi_4|\phi_3\rangle - 4\langle\phi_4|\phi_4\rangle = 0$$

$$\therefore \langle\psi_1|\psi_2\rangle = 0, \quad \langle\psi_1|\psi_1\rangle = 1$$

$$\Rightarrow -6i - i - 5\alpha - 4 = 0$$

$$-7i - 4 = 5\alpha$$

$$\alpha = \frac{-7i - 4}{5}$$

• Q. (v) •

What is the physical significance of Hamiltonian in Quantum Mechanics?

• Ans. •

In Quantum Mechanics

Page # 56

Topic : Physical significance
of Hamiltonian

• vi •

Define complementry and
correspondance principle?

• Answer •

Chapter # 2

Page # 76 (only definition)

Ilmi Quantum Mechanics.

• vii •

Write down physical
significance of Ladder
operators in Quantum
Mechanics.

• Answer •

Example # 5.16

Pg # 241, 242

Chapter # 5

Ilmi Quantum Physics.

~(viii)~.

In case of orbital momentum operator, can $\hat{L}^2$ and $\hat{L}_z$ both be simultaneously measured? Explain the reason.

~(Answer)~.

Topic 5.4

Page # 236 - 238

Chapter # 5

Ilmi Quantum Mechanics.

~(ix)~.

Compute $[\hat{S}_z, \hat{S}_x] = ?$

~(Answer)~.

Example # 8.1 (c)

Page # 305, 306

Chapter # 8

Ilmi Quantum Mechanics.

~(x)~.

Prove that

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle.$$

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where $\hat{a}$ is annihilation operator.

Answer

The annihilation operator $\hat{a}$ is defined as

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right)$$

To prove:-

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$$

Now

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

$$\therefore N = \hat{a}^\dagger \hat{a}$$

$$\hat{a}^\dagger|n\rangle = n|n\rangle$$

as

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

Section II

LONG QUESTIONS

Question # 02:-

~ a b ~

Explain Stern-Gerlach
Experiment in detail.

~ Answer ~

Topic 8.8 (complete)

Chapter # 8

Page # 317 - 319

Ilmi Quantum Mechanics.

~ e (b) ~

Given that in spherical
coordinates $L_z = -i\hbar \frac{\partial}{\partial \phi}$,
determine whether $\sin m\phi$ or
not $\sin m\phi$ and $e^{im\phi}$
are eigen function of
 L_z .

~ Answer ~

Example 5.8

Page # 236

Chapter # 5

Ilmi Quantum Mechanics.

Question # 3:-

Find the reflection and transmission coefficients R and T for 1-dimensional potential step for the case when $E > V_0$, where

$$V(x) = \begin{cases} 0, & \text{for } x < 0 \\ V_0, & \text{for } x \geq 0 \end{cases}$$

And also show that $R+T=1$.

~E(Answer)~

Topic 4.4 (till ~~special case~~ ^{Proof, $R+T=1$})

Page # 182, 183 and 189 (half)

Chapter # 4

Ilmi Quantum Mechanics.

Question # 4:-

Define the term central potential. Start with the

time independent Schrodinger wave equation, obtain an expression of radial wave function.

~ Answer ~

Central Potential:-

A central potential is a potential energy function that depends only on the distance r from a central point:

$$V(\vec{r}) = V(r).$$

$\Rightarrow$ It is spherically symmetric, meaning it is invariant under rotations. **Examples**

include the gravitational potential and the Coulomb potential.

Derivation of the Radial Wave Function:-

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Start with the time-dependent Schrödinger equation in 3D,

$$-\frac{\hbar^2}{2m} (\nabla^2 \psi(\vec{r})) + V(r) \psi(\vec{r}) = E \psi(\vec{r}).$$

Since the (potential) potential is central, use spherical coordinates (r, θ, ϕ) , and write the wave-function as:

$$\psi(r, \theta, \phi) = R(r) Y_l^m(\theta, \phi)$$

Now apply the Laplacian in spherical coordinates;

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \left(\frac{\partial^2 \psi}{\partial \phi^2} \right)$$

Substituting the separated form and simplifying, we obtain the radial equation

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D.

$$\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} + \left[\frac{2m}{\hbar^2} (E - V(r)) - \frac{l(l+1)}{r^2} \right] R = 0$$

Define $u(r) = r R(r)$, which simplifies the radial equation to:

$$\frac{d^2 u}{dr^2} + \left[\frac{2m}{\hbar^2} (E - V(r)) - \frac{l(l+1)}{r^2} \right] u = 0$$

This is standard radial Schrödinger equation

Boundary conditions for radial Schrödinger Equation's Solution:-

To find physically acceptable solution of radial equation $u(r)$, we impose boundary conditions that ensure

0] the wave-function is normalizable and physically meaningful.

1. Behaviour at $r = 0$;

- The wave-function must be "finite and single valued" at the "origin".

- Since $\psi(r) = rR(r)$, we get;

$$\psi(0) = 0$$

PUACP

- This avoids an infinite probability density at the origin.

- For small r , the dominant term in the radial equation is $\frac{l(l+1)}{r^2}$

So acceptable situation
behaves as

$$u(r) \propto r^{l+1}$$

2. Behaviour at $r \rightarrow \infty$:-

- The wavefunction must "vanish" at infinity to insure that the total probability of finding the particle in space is "finite".

- So, we get

$$\lim_{r \rightarrow \infty} u(r) = 0.$$

- For bound state (like in atoms), this leads to exponentially decaying solutions.